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Effect of dietary sodium restriction on renal outcome in patients with chronic kidney disease

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Citation

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Review question

This meta-analysis aims to evaluate the effects of different levels of dietary sodium intake on renal outcome in patients with chronic kidney disease (CKD).

Searches

All relevant randomized controlled trails (RCTs) and cohort studies will be identified and searched in following electronic databases without language restriction: Ovid MEDLINE SP (from 1946 to July 2017), EMBASE (from 1966 to July 2017), and the Cochrane Central Register of Controlled Trails(no date restriction). A search in www.clinicaltrials.gov will be used to identify the ongoing but unpublished studies about this field. We combined this strategy with a manual search of reference lists from identified trials and review articles.

Search strategy

https://www.crd.york.ac.uk/PROSPEROFILES/72395_STRATEGY_20170618.pdf

Types of study to be included

We will include RCTs and cohort studies that compared different levels of sodium intake for adult patients with CKD as well as studies in broader populations in which outcome data for subgroups with CKD could be disaggregated.

Condition or domain being studied

Chronic kidney disease

Participants/population

Inclusion criteria: Adults (no less than 18 years old) with CKD (as defined by KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of CKD) at all disease stages will included without restrictions on gender or race. Data from subgroups of participants with CKD within studies with broader inclusion criteria (e.g. people from the general population, people with diabetes, people with cardiovascular disease) were also included.
Exclusion criteria: Studies which do not include any people with chronic kidney disease.

Intervention(s), exposure(s)

Low-sodium diet

Comparator(s)/control

No intervention or different levels of dietary sodium intake

Context

None

Primary outcome(s)

If the data allow:

1) Kidney failure events: It will include more than 25% or 50% decrease in estimated glomerular filtration rate (eGFR), doubling of serum creatinine, end stage renal disease (ESRD);
2) Rate of change in the eGFR: The difference from baseline in eGFR divided by the number of years between creatinine measurements (ml/min/1.73m2 per year);
3) Change in proteinuria or albuminuria, including urinary protein excretion (UPE), urinary albumin excretion (UAE), protein to creatinine (PCR) and albumin to creatinine ratio (ACR);
4) Major cardiovascular events: It will include a composite including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death.

Secondary outcome(s)

1) 24-hour urinary sodium excretion;
2) Change in blood pressure;
3) All-cause mortality;
4) Change in markers of fluid overload (brain natriuretic peptide, N-terminal pro-brain natriuretic peptide, weight, bio-impedance analysis);
5) Adverse events: hypotensive episodes, undesirable change in blood lipids (low density lipoprotein, high-density lipoprotein);
6) Hospitalization.

Data extraction (selection and coding)

Data extraction will be carried out independently by two reviewers using standard data extraction forms. Disagreement will be resolved by consensus or by discussion with a third author. When detailed information that was needed for the analysis was not available, the original authors will be contacted to obtain the missing information through e-mail. Where multiple publications of one study exists, data will be extracted and grouped together as a single study with the most complete data.
The extracted data will include: study characteristics (study design, method of randomization, and withdrawals / dropouts); baseline patient characteristics (age, sex, race, country, BMI, sodium intake and its measurement, 24-hour urinary sodium excretion, mean proteinuria or albuminuria, eGFR, serum creatinine concentrations, blood lipids, blood pressure, diabetic status, cardiovascular events, smoking status, glycosylated hemoglobin, hemoglobin, antihypertensive drugs, antidiabetic drugs, different kidney disease diagnosis); change in eGFR; follow-up duration; outcomes: as listed in Primary outcomes and Secondary outcomes. When the required quantitative data is only provided in published figures, we will use g3 data software (http://www.frantz.fi/software/g3data.php) to extract exact numbers from figures.

Risk of bias (quality) assessment

For the RCTs, we will assess risk of bias for sequence generation, allocation concealment, blinding, selective reporting, incomplete outcome data and other sources of bias, and determined overall risk of bias based on predefined rules, utilizing the Cochrane Collaboration risk of bias tool. (reference: Higgins JP, Altman DG, et al. Cochrane Statistical Methods G. The Cochrane Collaboration's tool for assessing risk of bias in randomized trials. BMJ 2011;343:d5928.). Furthermore, the quality of RCT study will be quantified with Jadad scale. (reference: Jadad AR, Moore RA, Carroll D, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Control Clin Trials 1996;17:1– 12.). The quality of cohort study will be independently evaluated by each investigator using Newcastle-Ottawa quality assessment scale. (reference: Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010; 25:603–5.).

Strategy for data synthesis

The results of the data extraction and quality assessment for each study will be presented in structured tables. For dichotomous outcomes, relative risks (RR) and 95% Cis will be calculated from the numbers of events and participants at risk for events. For continuous outcomes, the mean difference (MD) will be used, or the standardized mean difference (SMD) if different scales were used. Data will be pooled using the random-effects model but the fixed-effect model will also be used to ensure robustness of the model chosen and susceptibility to outliers.
Heterogeneity will be analyzed using Cochrane’s Q and the I-squared statistic. We will explore potential

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heterogeneity by subgroup analysis and meta-regression.
Sensitivity analyses will be conducted to check the robustness of the effect of sodium-restricted treatment by excluding of trials with small sample size and low Jadad scores.
A two-sided p-value less than 0.05 was considered statistically significant and statistical analyses were performed using RevMan version 5.3 and STATA version 12.0.

Analysis of subgroups or subsets
If the data allow, subgroup analysis will be conducted by study design, follow-up duration, measures of sodium intake, different levels of sodium intake, difference in sodium intake between two groups, and baseline measures, including age, sex, BMI, eGFR, 24-hour urinary sodium excretion,mean serum creatinine, blood pressure, mean proteinuria or albuminuria, stage of CKD, presence of comorbidities (hypertension, diabetes and CVD), different kidney disease diagnosis (e.g. GN or Kidney Transplant Recipients) and whether RAAS blocker or other antihypertensive drug treatments.

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Details of any existing review of the same topic by the same authors
None

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions
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